Analysis of Building Frames (Using Kani's Method)

Gasper Kani, a German engineer, developed another distribution procedure based on slope deflection equations. This method is very useful for the analysis of multistorey frames. The greatest advantage of this method is, even if a mistake is committed in distribution in one of the cycles, it converges finally to the correct answer. Even today, many practising engineers who are not familiar with computer methods, use Kani's method for the analysis of 3 to 4 storey building frames.

This method is first explained for structures with fixed ends. Then, the modifications to handle simply supported and overhanging ends are discussed. Analysis of symmetric frames making use of the symmetry is also explained, after which an analysis of general frames is taken up.

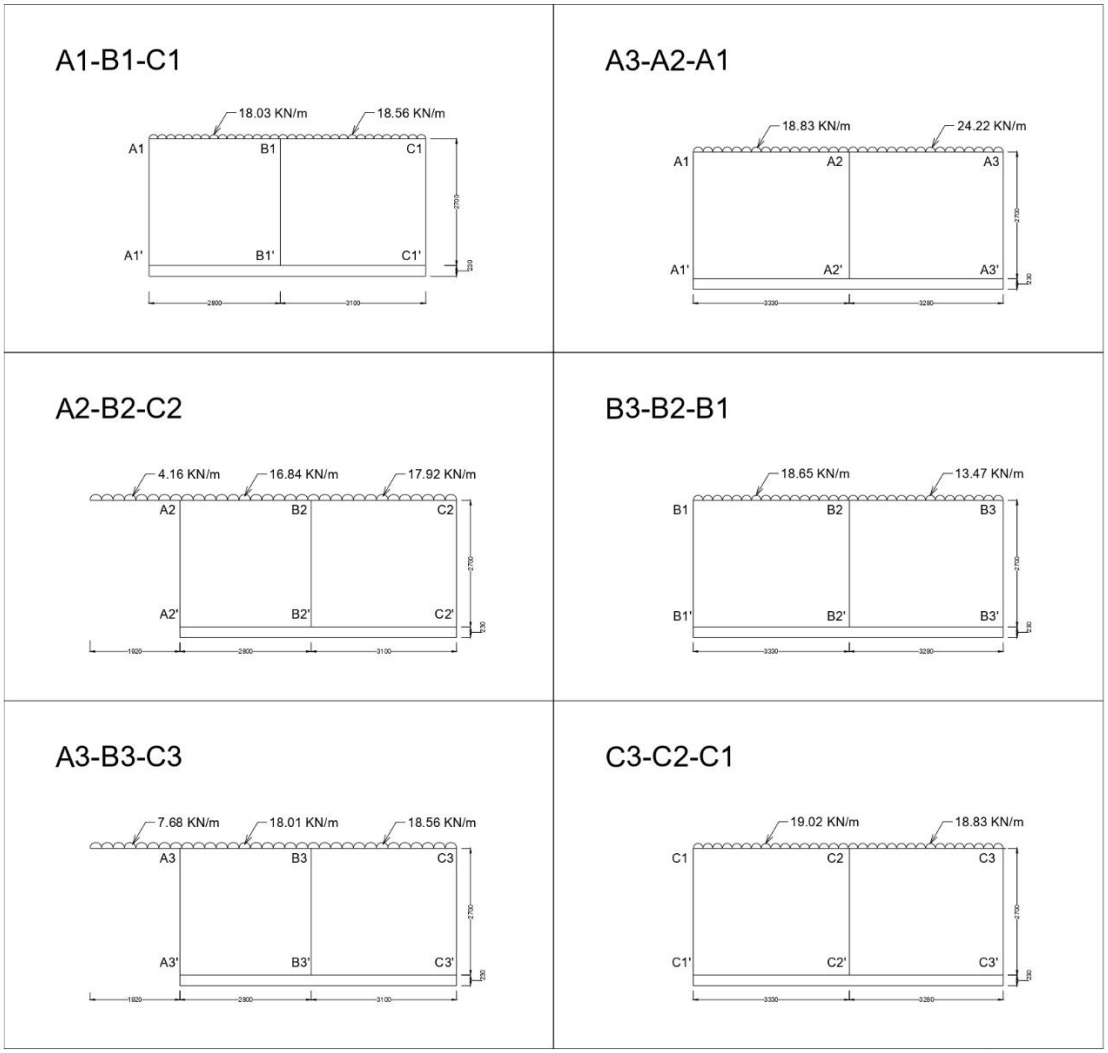


Figure 1: Beam Layout

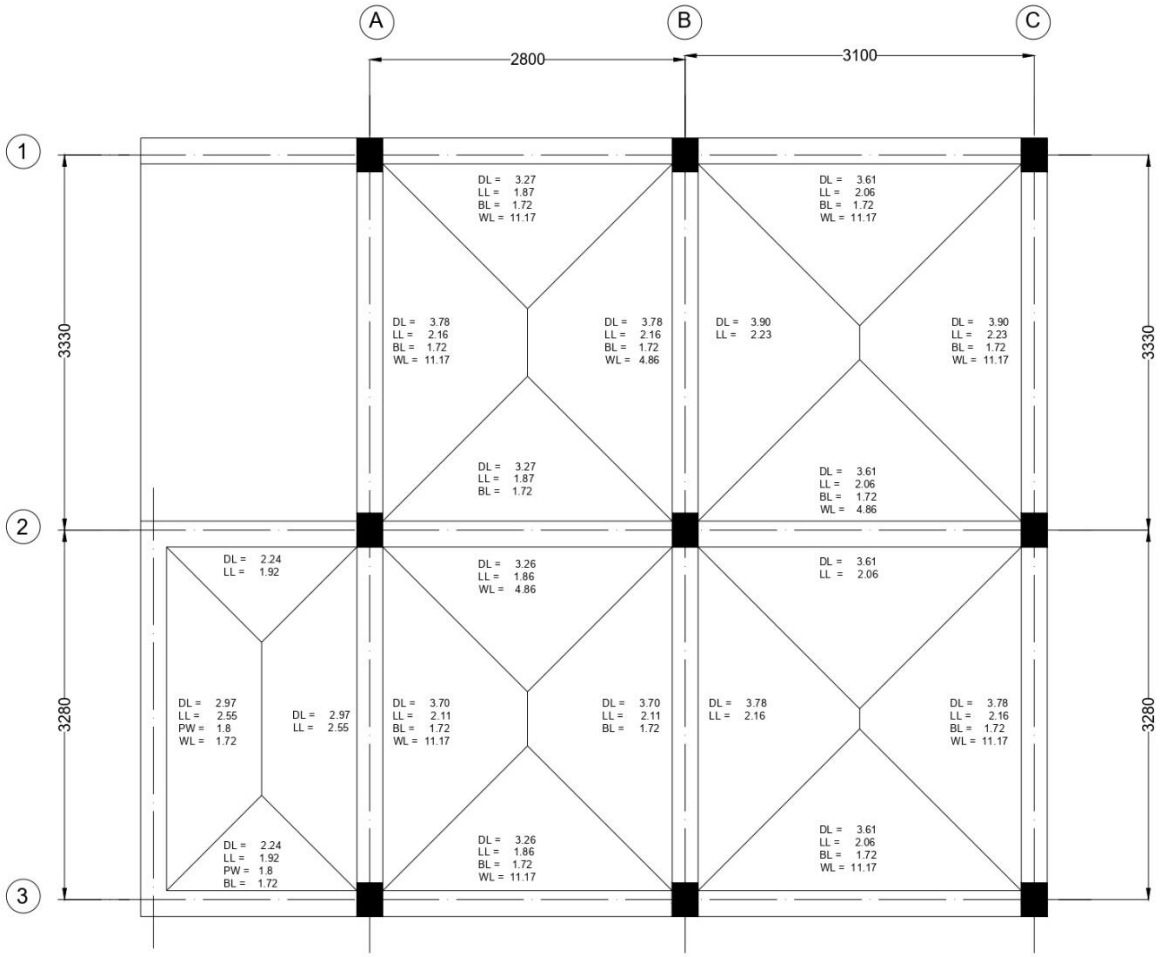
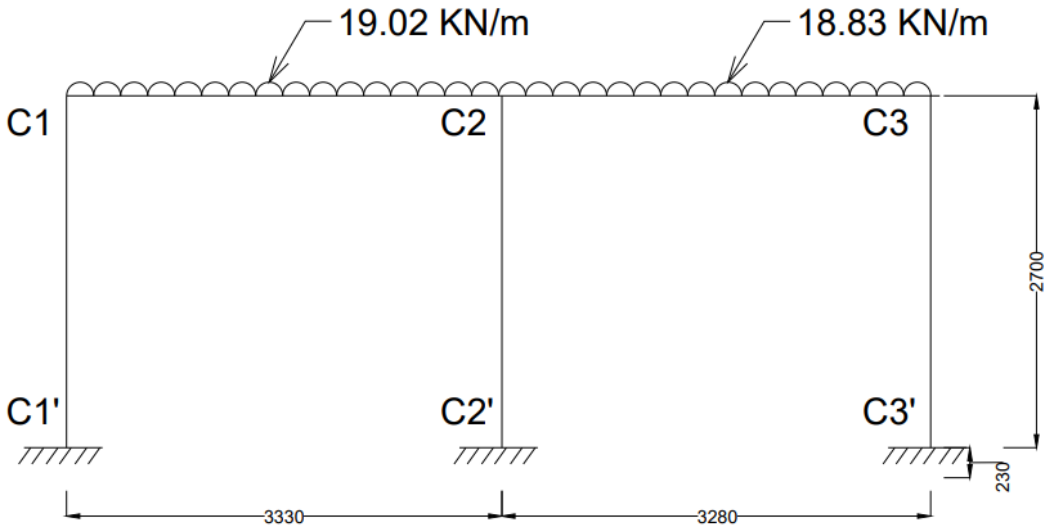


Figure 2: Frames.

Analysis of Frame C1-C2-C3



Step 1: Fixed End Moments

$MF_{C1C2} = \frac{-WL^2}{12} = \frac{-19.02 \times 3.33^2}{12}$ = -17.58 KN-m

$MF_{C2C1} = \frac{WL^2}{12} = \frac{19.02 \times 3.33^2}{12}$ = +17.58 KN-m

$MF_{C2C3} = \frac{-WL^2}{12} = \frac{-18.83 \times 3.28^2}{12}$ = -16.88 KN-m

$MF_{C3C2} = \frac{WL^2}{12} = \frac{18.83 \times 3.28^2}{12}$ = +16.88 KN-m

Fixed end moments in All the columns are zero.

i.e,

$MF_{C1C1'} = MF_{C1'C1} = MF_{C2C2'} = MF_{C2'C2} = MF_{C3C3'} = MF_{C3'C3} = 0$

Step 2: Rotation Factor $\left(\mathbf{R.F} = -\frac{1}{2}\left(\frac{k}{\Sigma k}\right)\right), \left(k = \frac{4EI}{L}\right)$

Joints	Members	Stiffness (k)	Σk	R.F
C1	C1C2	1.201EI	2.682EI	-0.224
	C1C1′	1.481EI		-0.276
C2	C2C1	1.201EI	3.902EI	-0.154
	C2C3	1.220EI		-0.156
	C2C2′	1.481EI		-0.190
C3	C3C2	1.220EI	2.701EI	-0.226
	C3C3′	1.481EI		-0.274

Stiffness (k) $\left(k = \left(\frac{4EI}{L}\right)\right)$

$K_{C1C2} = \left(\frac{4EI}{3.33}\right) = 1.201EI$

$K_{C1C1'} = \left(\frac{4EI}{2.7}\right) = 1.481EI$

$K_{C2C1} = \left(\frac{4EI}{3.33}\right) = 1.201EI$

$K_{C2C3} = \left(\frac{4EI}{3.28}\right) = 1.220EI$

$K_{C2C2'} = \left(\frac{4EI}{2.7}\right) = 1.481EI$

$K_{C3C2} = \left(\frac{4EI}{3.28}\right) = 1.220EI$

$K_{C3C3'} = \left(\frac{4EI}{2.7}\right) = 1.481EI$

Rotation Factor $\left(\mathbf{R.F} = -\frac{1}{2}\left(\frac{k}{\Sigma k}\right)\right)$

$R.F_{C1C2} = -\frac{1}{2}\left(\frac{1.201}{2.682EI}\right) = -0.224$

$R.F_{C1C1'} = -\frac{1}{2}\left(\frac{1.481EI}{2.682EI}\right) = -0.276$

$R.F_{C2C1} = -\frac{1}{2}\left(\frac{1.201EI}{3.902EI}\right) = -0.154$

$R.F_{C2C3} = -\frac{1}{2}\left(\frac{1.220EI}{3.902EI}\right) = -0.156$

$R.F_{C2C2'} = -\frac{1}{2}\left(\frac{1.481EI}{3.902EI}\right) = -0.190$

$R.F_{C3C2} = -\frac{1}{2}\left(\frac{1.220EI}{2.701EI}\right) = -0.226$

$R.F_{C3C3'} = -\frac{1}{2}\left(\frac{1.481EI}{2.701EI}\right) = -0.274$

Sum of Stiffness $\sum k$ at Joints

C1: 1.201EI + 1.481EI = 2.682EI

C2: 1.201EI + 1.220EI + 1.481EI = 3.902EI

C3: 1.220EI + 1.481EI = 2.701EI

Step 3: Displacement Factor $\left(\mathbf{D.F} = -\frac{3}{2}\left(\frac{k}{\Sigma k}\right)\right)$

Storey	Members	k	Σk	D.F
1	C1C1′	1.481EI	4.444EI	-0.5
	C2C2′	1.481EI		-0.5
	C3C3′	1.481EI		-0.5

Stiffness (k) $\left(k = \left(\frac{4EI}{L}\right)\right)$

$K_{C1C1'} = \left(\frac{4EI}{2.7}\right) = 1.481EI$

$K_{C2C2'} = \left(\frac{4EI}{2.7}\right) = 1.481EI$

$K_{C3C3'} = \left(\frac{4EI}{2.7}\right) = 1.481EI$

Displacement Factor $\left(\mathbf{D.F} = -\frac{3}{2}\left(\frac{k}{\Sigma k}\right)\right)$

$D.F_{C1C1'} = -\frac{3}{2}\left(\frac{1.481EI}{4.444EI}\right) = -0.5$

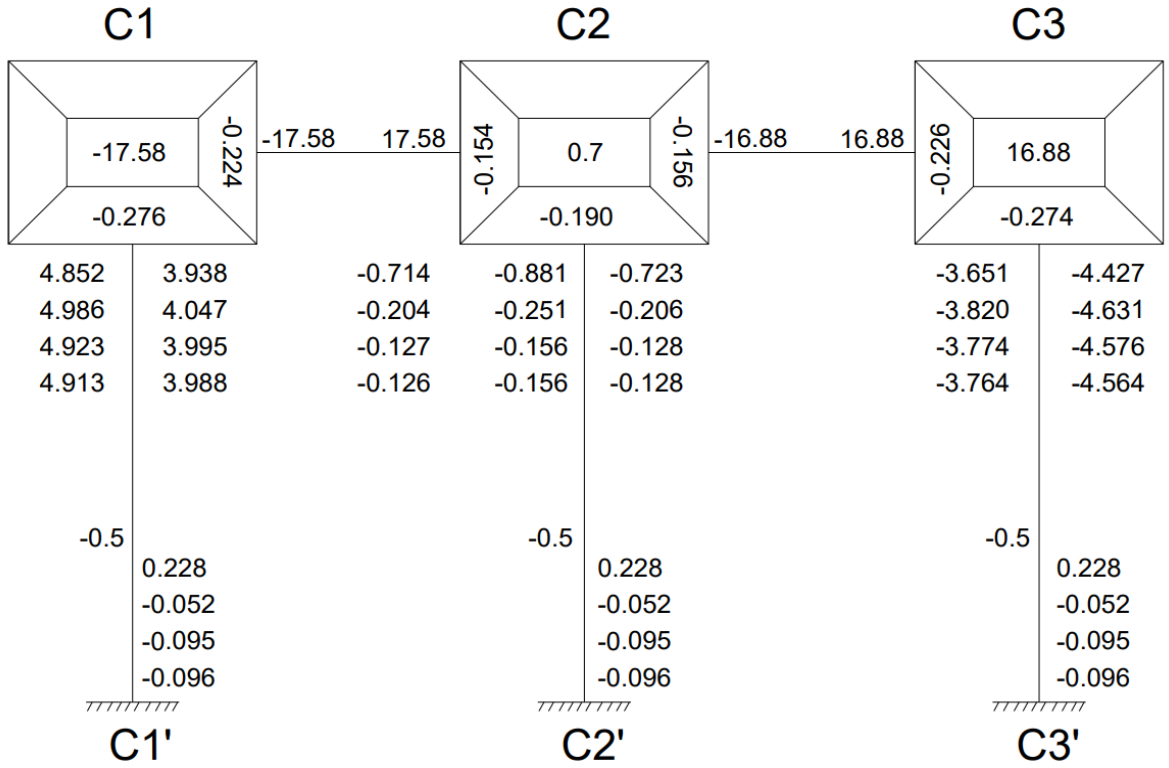
$D.F_{C2C2'} = -\frac{3}{2}\left(\frac{1.481EI}{4.444EI}\right) = -0.5$

$D.F_{C3C3'} = -\frac{3}{2}\left(\frac{1.481EI}{4.444EI}\right) = -0.5$

Sum of Stiffness $\sum k$ at Storey

$\sum k = 1.481EI + 1.481EI + 1.481EI = 4.444EI$

Rotation and Displacement Contribution



Cycle 1

Joint Rotation $(\boldsymbol{C.V.} = \boldsymbol{R.F} \times [\sum \boldsymbol{FEM} + \boldsymbol{NCV} + \boldsymbol{FCV}])$

M_{C1C2} = -0.224 × [-17.58 + 0 + 0] = 3.938

$M_{C1C1'}$ = -0.276 × [-17.58 + 0 + 0] = 4.852

M_{C2C1} = -0.154 × [0.7 + 3.938 + 0] = -0.714

M_{C2C3} = -0.156 × [0.7 + 3.938 + 0] = -0.723

$M_{C2C2'}$ = -0.190 × [0.7 + 3.938 + 0] = -0.881

M_{C3C2} = -0.226 × [16.88 + (-0.723) + 0] = -3.651

$M_{C3C3'}$ = -0.274 × [16.88 + (-0.723) + 0] = -4.427

Displacement Contribution

For All Columns $(\boldsymbol{R.V.} = \boldsymbol{D.F} \times [\sum \boldsymbol{Rotations\ Contribution\ at\ Columns}])$

$M_{C1C1'} = M_{C2C2'} = M_{C3C3'}$ = -0.5 × (4.852 + (-0.881) + (-4.427)) = 0.228

Cycle 2

Joint Rotation $(\boldsymbol{C.V.} = \boldsymbol{R.F} \times [\sum \boldsymbol{FEM} + \boldsymbol{NCV} + \boldsymbol{FCV}])$

M_{C1C2} = -0.224 × [-17.58 + 0.228 + (-0.714)] = 4.047

$M_{C1C1'}$ = -0.276 × [-17.58 + 0.228 + (-0.714)] = 4.986

M_{C2C1} = -0.154 × [0.7 + 0.228 + 4.047 + (-3.651)] = -0.204

M_{C2C3} = -0.156 × [0.7 + 0.228 + 4.047 + (-3.651)] = -0.206

$M_{C2C2'}$ = -0.190 × [0.7 + 0.228 + 4.047 + (-3.651)] = -0.251

M_{C3C2} = -0.226 × [16.88 + 0.228 + (-0.206)] = -3.820

$M_{C3C3'}$ = -0.274 × [16.88 + 0.228 + (-0.206)] = -4.631

Displacement Contribution

For All Columns $(\boldsymbol{R.V.} = \boldsymbol{D.F} \times [\sum \boldsymbol{Rotations\ Contribution\ at\ Columns}])$

$M_{C1C1'} = M_{C2C2'} = M_{C3C3'}$ = -0.5 × (4.986 + (-0.251) + (-4.631)) = -0.052

Cycle 3

Joint Rotation $(\boldsymbol{C.V.} = \boldsymbol{R.F} \times [\sum \boldsymbol{FEM} + \boldsymbol{NCV} + \boldsymbol{FCV}])$

M_{C1C2} = -0.224 × [-17.58 + (-0.052) + (-0.204)] = 3.995

$M_{C1C1'}$ = -0.276 × [-17.58 + (-0.052) + (-0.204)] = 4.923

M_{C2C1} = -0.154 × [0.7 + (-0.052) + 3.995 + (-3.820)] = -0.127

M_{C2C3} = -0.156 × [0.7 + (-0.052) + 3.995 + (-3.820)] = -0.128

$M_{C2C2'}$ = -0.190 × [0.7 + (-0.052) + 3.995 + (-3.820)] = -0.156

M_{C3C2} = -0.226 × [16.88 + (-0.052) + (-0.128)] = -3.774

$M_{C3C3'}$ = -0.274 × [16.88 + (-0.052) + (-0.128)] = -4.576

Displacement Contribution

For All Columns $(\boldsymbol{R.V.} = \boldsymbol{D.F} \times [\sum \boldsymbol{Rotations\ Contribution\ at\ Columns}])$

$M_{C1C1'} = M_{C2C2'} = M_{C3C3'}$ = -0.5 × (4.923 + (-0.156) + (-4.576)) = -0.095

Cycle 4

Joint Rotation $(\boldsymbol{C.V.} = \boldsymbol{R.F} \times [\sum \boldsymbol{FEM} + \boldsymbol{NCV} + \boldsymbol{FCV}])$

M_{C1C2} = -0.224 × [-17.58 + (-0.095) + (-0.127)] = 3.988

$M_{C1C1'}$ = -0.276 × [-17.58 + (-0.095) + (-0.127)] = 4.913

M_{C2C1} = -0.154 × [0.7 + (-0.095) + 3.988 + (-3.774)] = -0.126

M_{C2C3} = -0.156 × [0.7 + (-0.095) + 3.988 + (-3.774)] = -0.128

$M_{C2C2'}$ = -0.190 × [0.7 + (-0.095) + 3.988 + (-3.774)] = -0.156

M_{C3C2} = -0.226 × [16.88+ (-0.095) + (-0.128)] = -3.764

$M_{C3C3'}$ = -0.274 × [16.88 + (-0.095) + (-0.128)] = -4.564

Displacement Contribution

For All Columns $(\boldsymbol{R.V.} = \boldsymbol{D.F} \times [\sum \boldsymbol{Rotations\ Contribution\ at\ Columns}])$

$M_{C1C1'} = M_{C2C2'} = M_{C3C3'}$ = -0.5 × (4.913 + (-0.156) + (-4.564)) = -0.096

Step 4: Final End Moments

M_{ij} = FEM + 2×M_{ij} (Near end Rotation) + M_{ji} (Far end Rotation) +M″ (Displacement)

Beams (No Sway M″= 0)

M_{C1C2} = -17.58 + 2(3.988) + (-0.126) = -9.73

M_{C2C1} = 17.58 + 2(-0.126) + 3.988 = 21.31

M_{C2C3} = -16.88 + 2(-0.128) + (-3.764) = -20.9

M_{C3C2} = 16.88 + 2(-3.764) + (-0.128) = 9.22

Column (With Sway M″= (-0.096))

$M_{C1C1'}$ = 0 + 2(4.913) + 0 + (-0.096) = 9.73

$M_{C1'C1}$ = 0 + 2(0) + 4.913 + (-0.096) = 4.82

$M_{C2C2'}$ = 0 + 2(-0.156) + 0 + (-0.096) = -0.41

$M_{C2'C2}$ = 0 + 2(0) + (-0.156) + (-0.096) = -0.25

$M_{C3C3'}$ = 0 + 2(-4.564) + 0 + (-0.096) = -9.22

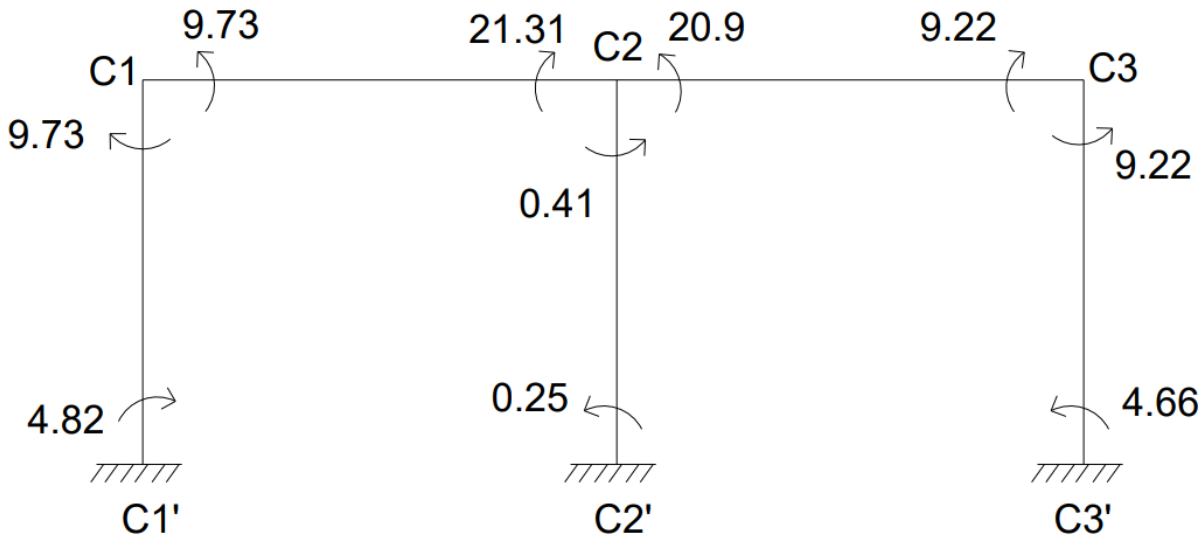
$M_{C3'C3}$ = 0 + 2(0) + (-4.564) + (-0.096) = -4.66

Equilibrium Check

Joint C1 = M_{C1C2} + $M_{C1C1'}$ = -9.73 + 9.73 = 0

Joint C2 = M_{C2C1} + $M_{C2C3'}$ + $M_{C2C2'}$ = 21.31 + (-20.9) + (-0.41) = 0

Joint C3 = M_{C3C2} + $M_{C3C3'}$ = 9.22 + (-9.22) = 0

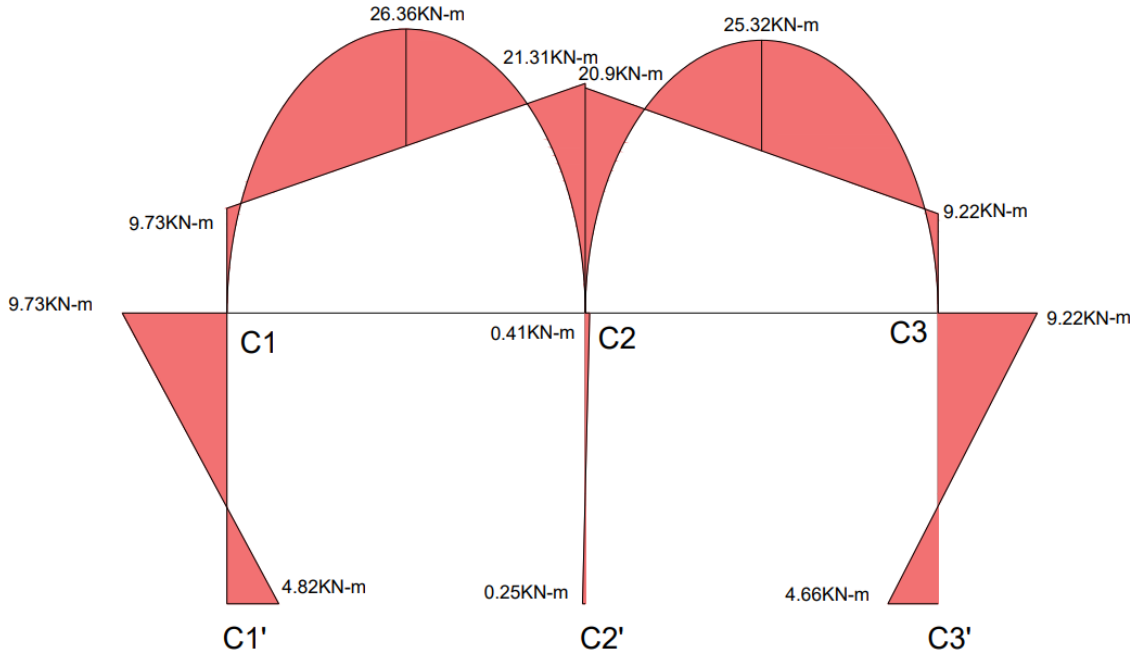


CALCULATION BENDING MOMENT

$BM_{C1C2} = \frac{WL^2}{8} = \frac{19.02 \times 3.33^2}{8} = 26.36KN\text{-}m$

$BM_{C2C3} = \frac{WL^2}{8} = \frac{18.83 \times 3.28^2}{8} = 25.32KN\text{-}m$

BENDING MOMENT DIAGRAM



CALCULATION OF REACTIONS

SPAN C1-C2

Length (L) = 3.33m, UDL (W) = 19.02KN/m

Simply Supported Reactions:

$R_{total} = \frac{WL}{2} = \frac{19.02 \times 3.33}{2} = 31.67KN$

Reaction Due to End Moments:

$R_{moments} = \frac{M_{C2C1} - M_{C1C2}}{L} = \frac{21.31 - 9.73}{3.33} = 3.48KN$

Final Reactions:

$R_{C1} = R_{total} - R_{moments} = 31.67 - 3.48 = 28.19KN$

$R_{C2(Left)} = R_{total} + R_{moments} = 31.67 + 3.48 = 35.15KN$

SPAN C2-C3

Length (L) = 3.28m, UDL (W) = 18.83KN/m

Simply Supported Reactions:

$R_{total} = \frac{WL}{2} = \frac{18.83 \times 3.28}{2} = 30.88KN$

Reaction Due to End Moments:

$R_{moments} = \frac{M_{C2C3} - M_{C3C2}}{L} = \frac{20.9 - 9.22}{3.28} = 3.56KN$

Final Reactions:

$R_{C2(Right)} = R_{total} + R_{moments} = 30.88 + 3.56 = 34.44KN$

$R_{C3} = R_{total} - R_{moments} = 30.88 - 3.56 = 27.32KN$

Total Reactions:

Reaction at Column C1 (R_{C1}) = 28.19KN

Reaction at Column C2 (R_{C2}) = $R_{C2(Left)} + R_{C2(Right)} = 35.15 + 34.44 = 69.59KN$

Reaction at Column C3 (R_{C3}) = 27.32KN

